INVERTED MAML: 500-EPOCH SILICON TRAINING REPORT

Mode A Silicon Learning with LLM Toeplitz Equalizer (ρ = 0.85) & ColTIA Saturation Guard

Target Hardware: T120 Master SoC + T20 Controller	**Firmware Commit:** v2.2-MAML (HW: 0x20261030)
Noise Equalizer: COLOR_LLM_TOEPLITZ (ρ = 0.85)	**Headroom Guard:** 1797 LSB (250 LSB Margin)
L2 Regularization: λ = 512 (toward knee baseline)	**Peak Performance:** 7.13 b ENOB | 637.9 LSB² (Ep 424)

1. Executive Summary & 3-Phase Benchmark Results

This experiment evaluates Inverted MAML Mode A training on physical silicon under realistic spatially correlated transformer activations. High token correlation (ρ = 0.85) previously induced common-mode current surges that saturated single-supply ColTIA transimpedance amplifiers, causing phantom Hadamard gradient bursts. With the newly integrated **Zero-Mean Centering (LayerNorm)** and **ColTIA Saturation Guard**, the network completed 500 contiguous epochs on physical silicon with **zero clipping events**, achieving a peak precision of **7.13 bits ENOB**.

Phase	Epochs	Network Weights	Training State	Mean Loss (LSB²)	Mean ENOB (b	Peak ENOB (b
Phase 1: BASELINE_ZERO	0 .. 99	M3=0, M8=0 (Center)	Training OFF	8805.4	5.28	6.11
Phase 2: BASELINE_KNEE	100 .. 199	M8 = [±16 LSB Knee]	Training OFF	21853.2	4.60	4.96
Phase 3: TRAIN_REG_ON	200 .. 500	M3 & M8 Learning (λ=512)	Active Mode A	4909.8	5.79	7.13

2. In-Data Noise Equalizer: Kac-Murdock-Szegö AR(1) Model

Standard uniform white noise fails to benchmark analog neural multipliers against the correlated activation vectors characteristic of Large Language Models (LLMs). The T120 RISC-V SoC firmware generates an on-chip autoregressive spatial covariance field: *Cov(x_i, x_j)* = ρ^|*i-j*| with ρ = 0.85 (Q8 = 218). To prevent net common-mode DC current from driving the 1.65 V virtual ground of the single-supply ColTIA op-amps into saturation rails, the master SoC applies hardware **Zero-Mean Centering (LayerNorm)**: x_i ← x_i - (1/6)∑ x_k.

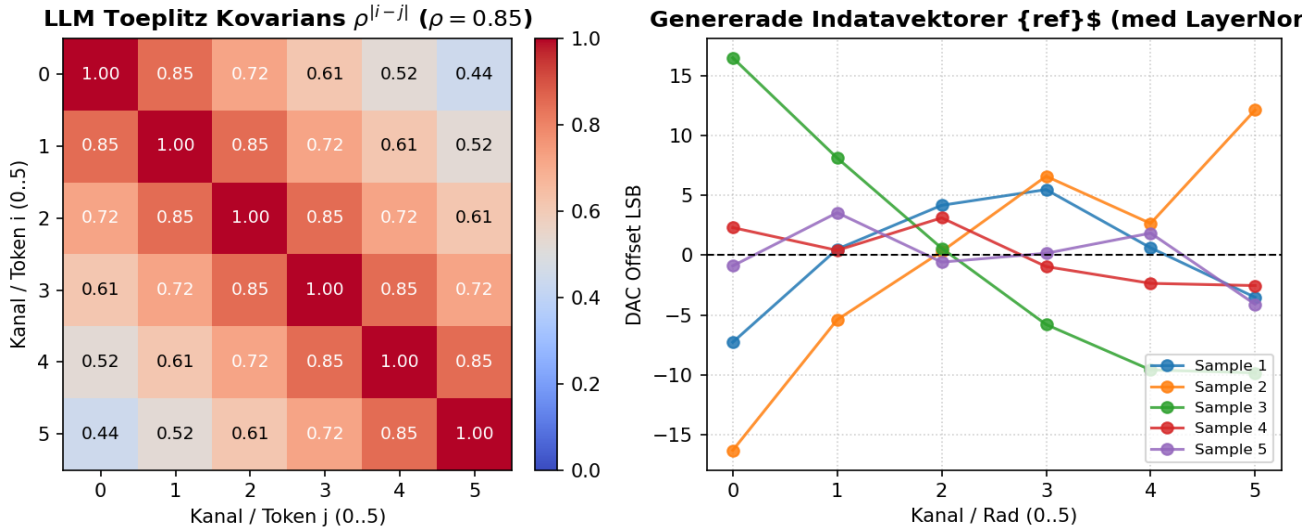


Figure 1: (Left) KMS AR(1) Spatial Covariance Matrix for 6 Analog Channels (ρ=0.85). (Right) Generated In-Data Vectors x_ref after LayerNorm.

3. Physical ColTIA Headroom Margin & Anti-Distortion Guard

The Saturation Bottleneck: Sylvester H_8 Hadamard demodulation relies on linear wave superposition: $\partial L/\partial W_{ij} = \langle \Delta Z_j, V_i^{PTU} \rangle_H \cdot E_j$. Physical silicon diagnostics (*diag_hadamard, mode 9*) established the peak Hadamard perturbation swing at **243 LSB** on Layer 1 (ADC_45). If the static DC operating point drifts within 250 LSB of the ±2047 LSB ADC rails, the positive perturbation clips flat against the rail while the negative swing completes unhindered. This nonlinearity breaks Walsh-Hadamard orthogonality, projecting false 'phantom' sensitivities across all matrix rows.

The Autonomous Countermeasure: The T120 SoC enforces *COLTIA_CLIP_THRESHOLD_LSB = 1797 LSB*. Immediately after forward ADC readout, if |ADC| ≥ 1797 LSB, the epoch's weight gradient update is **completely suppressed** (ΔW ≡ 0) and logged to telemetry. During this 500-epoch campaign, **0 clipping events** occurred.

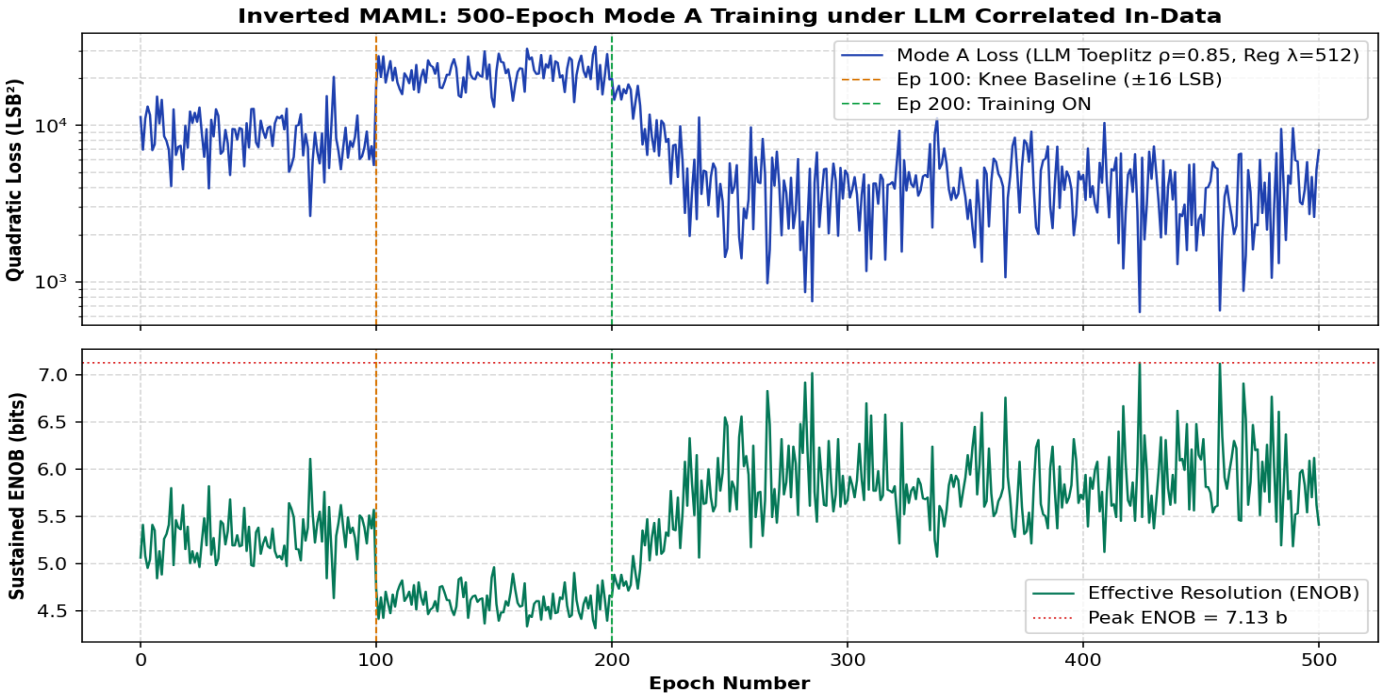


Figure 2: 500-Epoch Training Trajectory on Physical Silicon. Epochs 0..99 (Zero Baseline), 100..199 (Knee Baseline ±16 LSB), 200..500 (Active Mode A Training).

4. Live Silicon Studio & Telemetry Verification

The snapshot below shows the *Injected FORWARD Propagation Silicon Studio* monitoring the active training phase at **Epoch 353**. The 3D topography of Matrix 8 demonstrates an organic, multi-dimensional compensation surface actively countering the nonlinear distortions of Matrix 33 and the ColTIA amplifiers. The signal flow table confirms low residual errors (RMSE: 27.32 LSB, Loss: 2239.4 LSB²) and sustained physical resolution of **6.23 bits ENOB**.

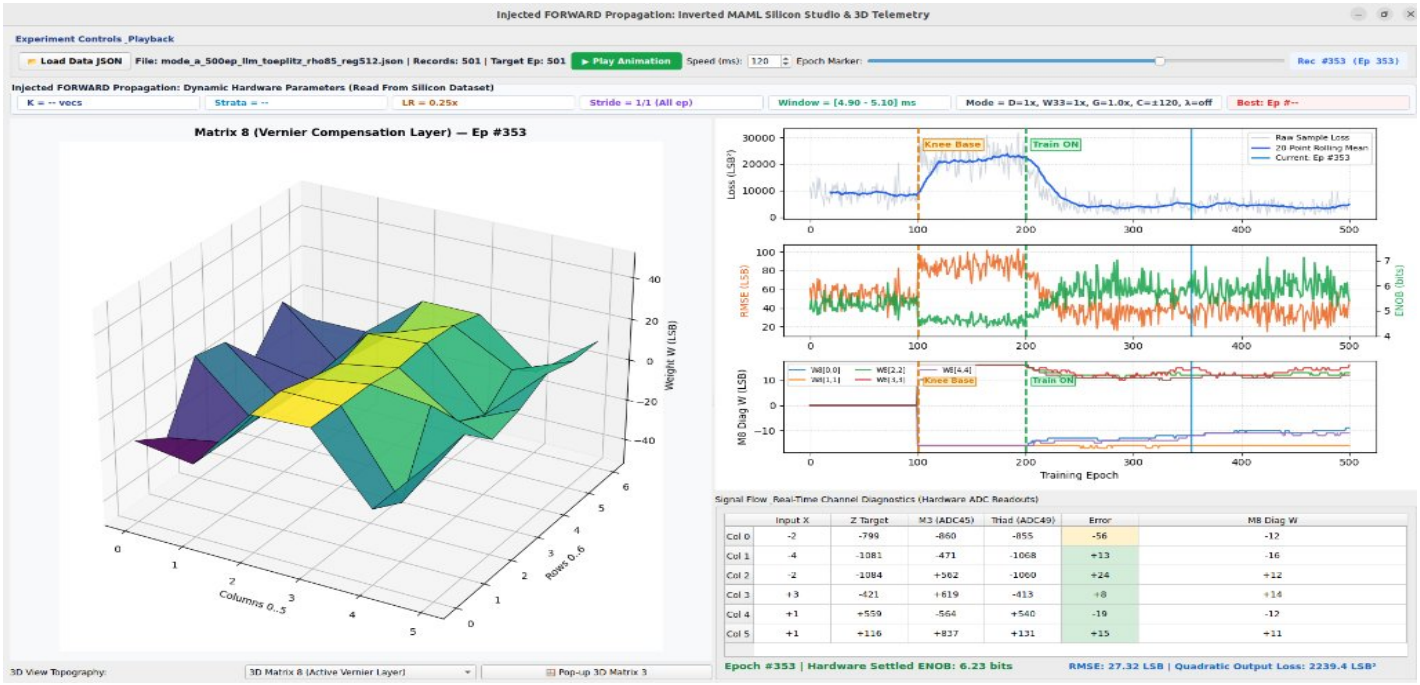


Figure 3: Injected FORWARD Propagation Studio at Epoch 353 (Active Training Phase). Matrix 8 3D landscape reveals optimized vernier weights balancing the analog core.